

Health and Fitness Tracking System

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1. Project Title

AI-Powered Health and Fitness Tracking System

2. Project Overview

The Health and Fitness Tracking System is an AI-powered no-code application that enables users to log workouts, track calories burned, monitor daily steps, and analyze weekly fitness progress.

The system automates:

- User management
- Workout logging
- Weekly goal tracking
- Report generation
- AI-based health suggestions
- Visual fitness analytics

The backend is powered by Airtable and automation tools like Make and n8n, while Gemini AI generates visual health reports.

3. Objectives

- Allow users to log fitness activities
- Track calories and step count
- Automatically manage user records
- Calculate weekly goal progress
- Generate automated fitness reports

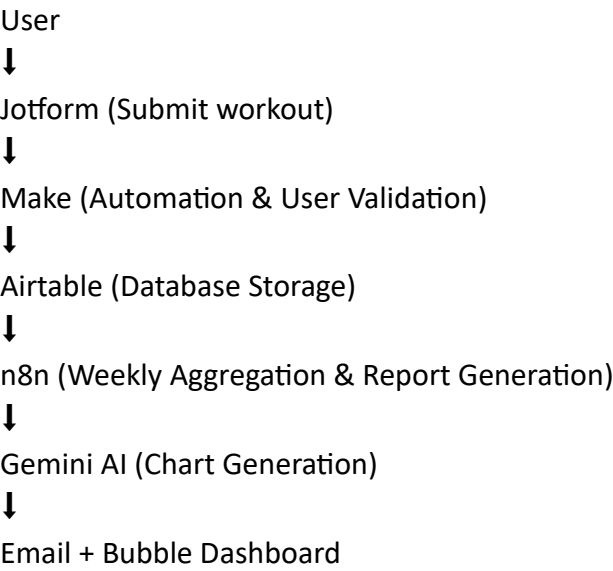
- Provide AI-based improvement suggestions
- Display visual analytics dashboards

 4. Tools Used

Tool	Purpose
Jotform	Collect workout data from users
Airtable	Backend database storage
Make	Automation & user creation logic
n8n	Weekly report generation workflow
Gemini AI (Nanobanana)	Generate fitness charts & visual analytics
Bubble	Frontend fitness tracking application
Notion AI	Project documentation

 5. System Architecture

System Flow:



6. Database Design (Airtable)

<https://airtable.com/appldQknnzCeQEgBQ/tblkHOZXQgKLanoMN/viwc0MQthIYwBShhv?blocks=hide>

Table 1: Users

Fields:

- Name
 - Email
 - Age
 - Weight
 - Height
 - Daily Step Goal
 - Weekly Calorie Goal
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Table 2: Workout Logs

Fields:

- User (Linked to Users)
- Date
- Workout Type
- Duration (Number)
- Calories Burned (Number)
- Steps (Number)

Fitness Tracker System

Grid view

1 hidden field

	A Name	Date	Workout Type	Duration	Calories Burned	Steps	Email
1	Ripe Sharma	2/10/2024	Strength	45.0	750.0	8,700.0	ripe@gmail.com
2	Gauthan M	2/10/2024	Cardio	35.0	550.0	5,500.0	gauthan@gmail.com
3	Sunil	2/11/2024	Yoga	30.0	450.0	5,000.0	sunil@gmail.com
4	Sunil	2/12/2024	Yoga	75.0	600.0	5,400.0	sunil@gmail.com

25 AM

http://fitnessjotform.com/2642/12376346

Full Name *

Sunil

Email

sunil@gmail.com

Workout Type

Yoga

Duration

75

Calories Burned

Jotform

Work with Copilot

Relationship:

Workout Logs → Linked to → Users (One-to-Many Relationship)

7. Automation Logic (Make)

<https://eu1.make.com/scenarios/4449624/edit>

Purpose:

Automatically create or validate user before logging workout.

Workflow:

1. Watch Jotform Submission
2. Search User in Airtable (by Email)
3. Router:
 - If User Exists → Create Workout Log
 - If User Not Found → Create User → Create Workout Log

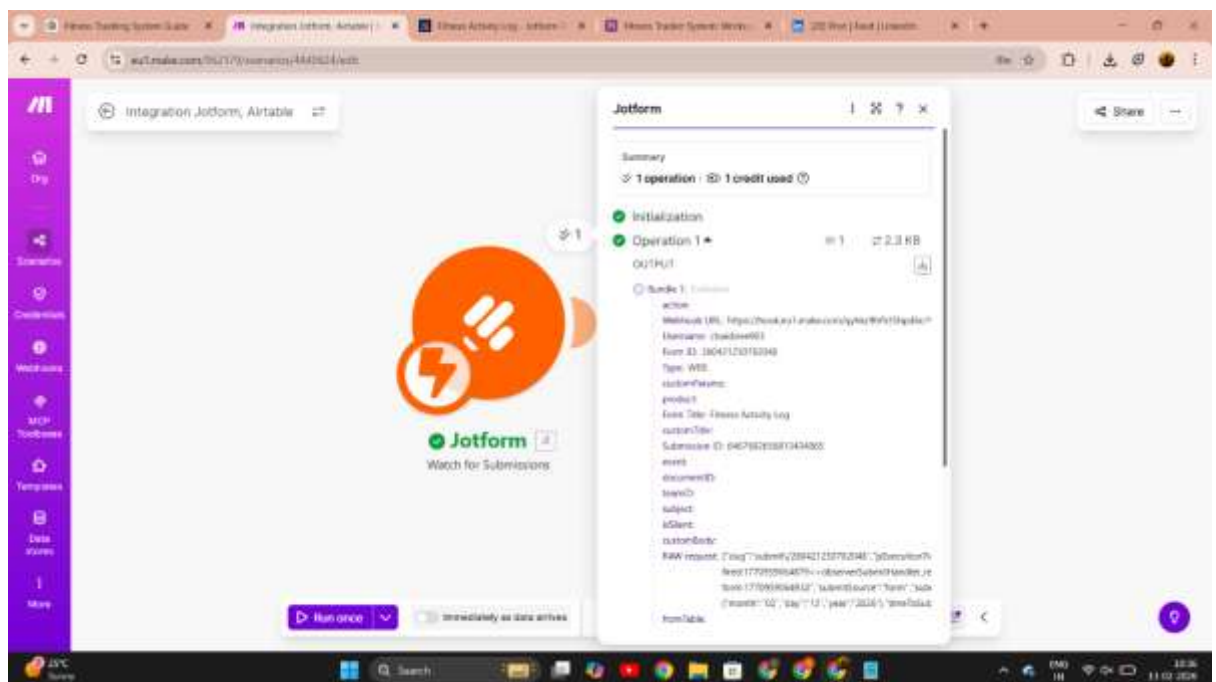
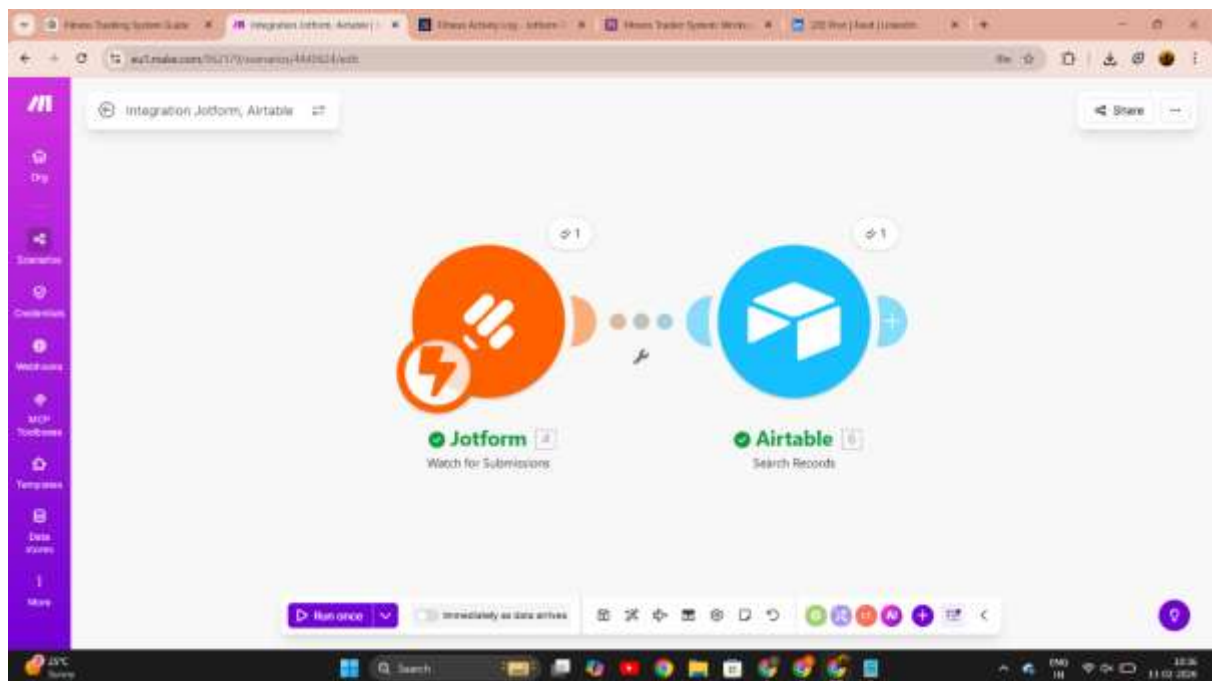
Key Logic:

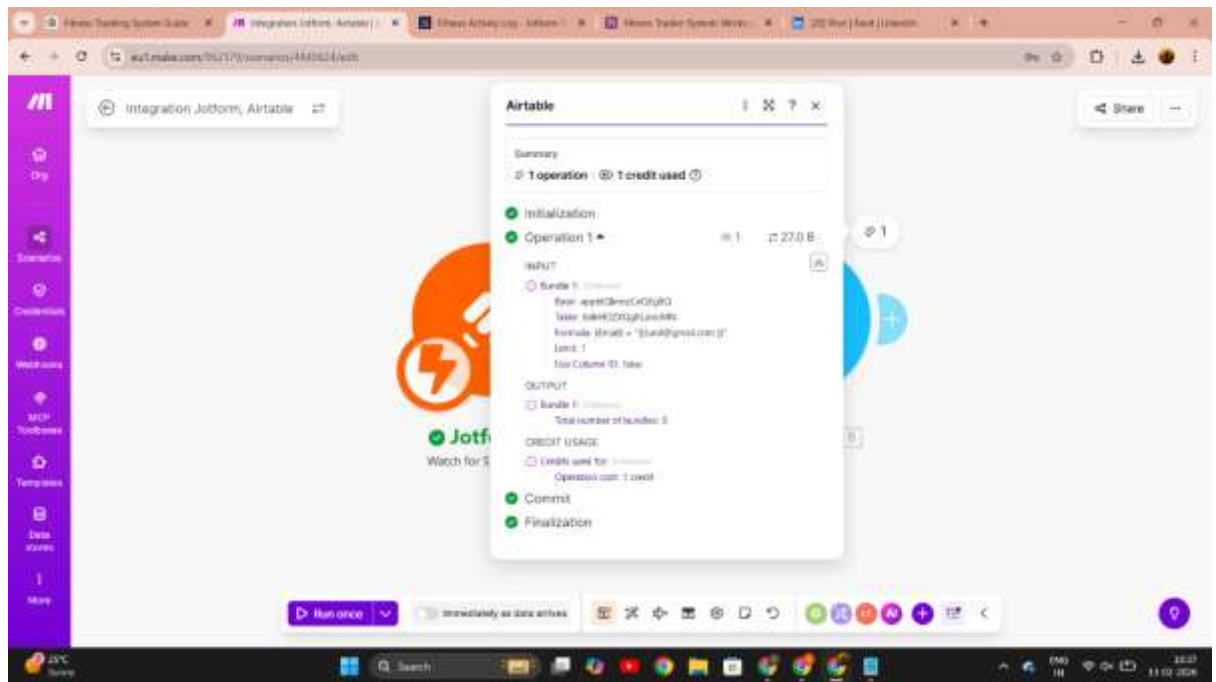
- If total search results > 0 → Existing user

- If total search results = 0 → New user creation

This ensures:

- No duplicate users
- Clean relational database
- Automatic backend updates





8. Weekly Report Automation (n8n)

Trigger:

Cron node → Every Sunday

Steps:

1. Fetch weekly workout records
2. Calculate:
 - Total Calories
 - Total Steps
 - Total Workouts
 - Goal Completion %
3. Generate summary text
4. Send data to Gemini AI
5. Receive generated charts
6. Send email report



9. AI Integration (Gemini AI)

Gemini AI generates:

- Bar chart (Weekly Calories)
- Line graph (Daily Steps)
- Pie chart (Goal Completion)

Example Prompt:

"Generate a modern fitness dashboard showing weekly calories, steps, and goal percentage using blue and green theme."

This enhances:

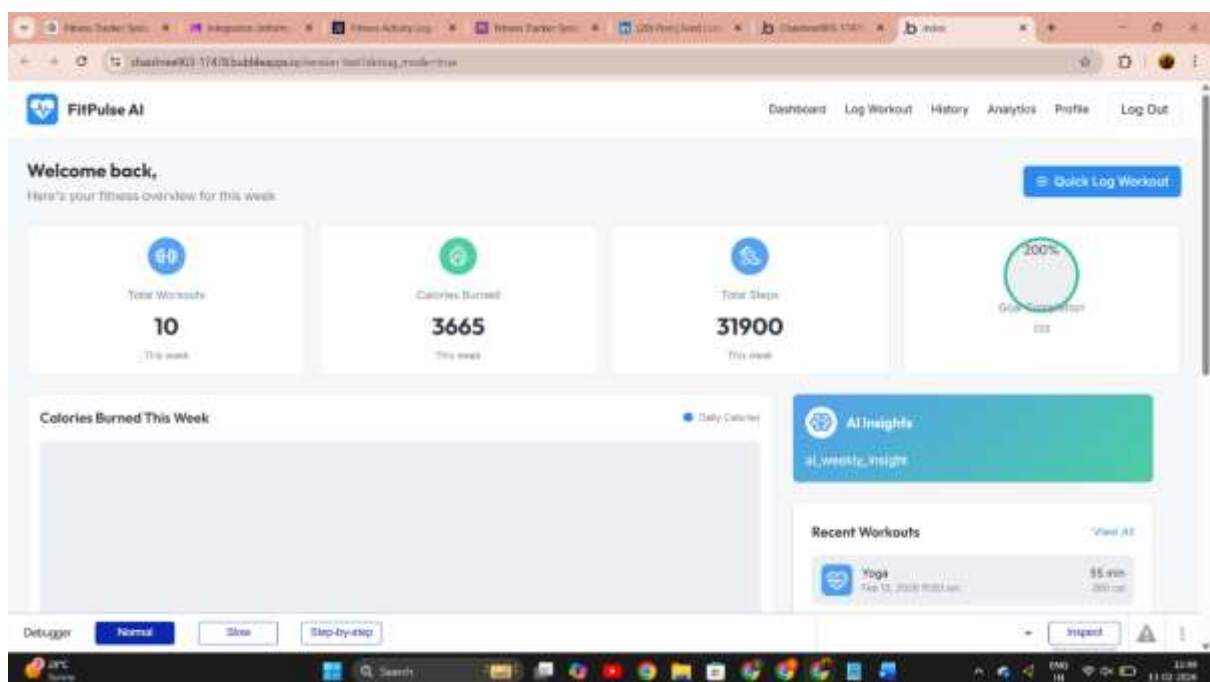
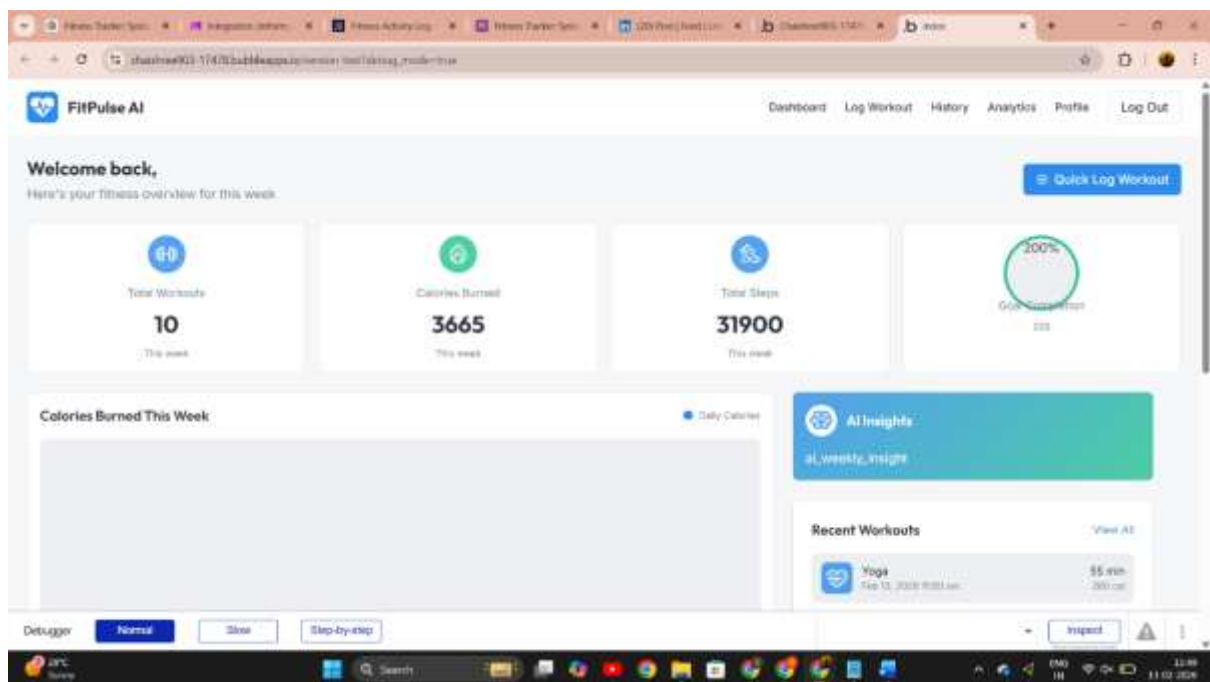
- Visual analytics
- Professional reporting
- AI-powered insights

10. Frontend (Bubble AI)

Bubble application provides:

- User Login
- Workout Log Form
- Weekly Dashboard
- Progress Bar
- Report Section
- AI Suggestions Section

The frontend connects to Airtable backend.



11. AI Suggestion Logic

If user goal completion < 80%:

System suggests:

- Increase workout duration
- Add 1 HIIT session
- Improve daily step count

Example AI Output:

"You are 15% away from your weekly goal. Adding one 30-minute cardio session can help you achieve it."

12. Sample Weekly Report Output

User: Riya Sharma

Workouts Completed: 4

Total Calories: 2200 kcal

Total Steps: 18,700

Goal Completion: 85%

Recommendations:

- Add one strength training session
- Increase daily step target by 1000
- Maintain current consistency

13. Data Validation & Error Handling

- Number conversion handled in Make
 - Duplicate user prevention logic
 - Linked record integrity maintained
 - Automatic type casting for numeric fields
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14. Key Features

- ✓ Automated user management
 - ✓ Relational database design
 - ✓ Weekly performance analytics
 - ✓ AI-generated fitness reports
 - ✓ Smart improvement suggestions
 - ✓ Fully no-code architecture
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15. Advantages of the System

- No manual backend updates required
 - Scalable for multiple users
 - Professional automation architecture
 - AI-powered analytics
 - Real-time workout tracking
 - Modular & extendable design
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16. Future Enhancements

- Integration with wearable devices
 - Real-time step tracking
 - Mobile app deployment
 - Advanced AI health predictions
 - Diet and nutrition tracking module
 - Social leaderboard system
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17. Conclusion

The Health and Fitness Tracking System successfully integrates no-code tools and AI technologies to automate fitness tracking, goal monitoring, and intelligent reporting.

By combining Jotform, Airtable, Make, n8n, and Gemini AI, the system demonstrates a scalable and professional approach to backend automation and AI-driven health analytics.

This project reflects practical implementation of:

- Workflow automation
- Relational database design
- AI integration
- Backend validation logic
- Real-world SaaS architecture